Tech Employment in the Age of AI: Milestone 3 Time Series Analysis

We Came, We Saw, We Modelled

Jensen Harvey (group leader), Kaitlyn Chou, and Emily Friedman

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1 Setup and Data Loading

```r
library(tidyverse)
library(lubridate)
library(scales)
library(forecast)
library(tseries)
library(zoo)
library(ggplot2)
library(patchwork)
library(knitr)
library(kableExtra)
```

```r
df <- read_csv("/Users/jensen.harvey/Desktop/data science/project2/Tech_layoffs.csv")

glimpse(df)
```

```
## Rows: 1,745
## Columns: 18
## $ Nr                          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,~
## $ Company                     <chr> "Tamara Mellon", "HopSkipDrive", "Panda Sq~
## $ Location_HQ                 <chr> "Los Angeles", "Los Angeles", "San Francis~
## $ Region                      <chr> "other", "other", "San Francisco Bay Area"~
## $ USState                     <chr> "California", "California", "California", ~
## $ Country                     <chr> "USA", "USA", "USA", "USA", "USA", "USA", ~
## $ Continent                   <chr> "North America", "North America", "North A~
## $ Laid_Off                    <dbl> 20, 8, 6, 16, 130, 4, 50, 30, 784, 375, 15~
## $ Date_layoffs                <date> 2020-03-12, 2020-03-13, 2020-03-13, 2020-~
## $ Percentage                  <dbl> 40, 10, 75, 100, 22, 20, 50, 50, 98, 15, 1~
## $ Company_Size_before_Layoffs <dbl> 50, 80, 8, 16, 591, 20, 100, 60, 800, 2500~
## $ Company_Size_after_layoffs  <dbl> 30, 72, 2, 0, 461, 16, 50, 30, 16, 2125, 6~
## $ Industry                    <chr> "Retail", "Transportation", "Consumer", "S~
## $ Stage                       <chr> "Series C", "Unknown", "Seed", "Seed", "Se~
## $ Money_Raised_in__mil        <dbl> 90, 45, 1, 6, 79, 4, 17, 7, 120, 1600, 280~
## $ Year                        <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, ~
## $ latitude                    <dbl> 34.05369, 34.05369, 37.77926, 30.27113, 39~
## $ longitude                   <dbl> -118.24277, -118.24277, -122.41933, -97.74~
```

```r
df <- df %>%
  mutate(
    YearMonth   = floor_date(Date_layoffs, "month"),
    Month       = month(Date_layoffs, label = TRUE, abbr = TRUE),
```

```r
    Laid_Off    = as.numeric(Laid_Off),
    Percentage  = as.numeric(Percentage)
  ) %>%
  filter(!is.na(Date_layoffs), !is.na(Laid_Off))

cat("Date range:", as.character(min(df$Date_layoffs)), "to",
    as.character(max(df$Date_layoffs)), "\n")
```

```
## Date range: 2020-03-12 to 2025-12-28
```

```r
cat("Total records:", nrow(df), "\n")
```

```
## Total records: 1745
```

```r
cat("Total employees laid off:", comma(sum(df$Laid_Off, na.rm = TRUE)), "\n")
```

```
## Total employees laid off: 581,705
```

---

2 Time Series Construction and Train/Test Split

2.1 Monthly Aggregation

```r
monthly_layoffs <- df %>%
  group_by(YearMonth) %>%
  summarise(Total_Laid_Off = sum(Laid_Off, na.rm = TRUE), .groups = "drop") %>%
  arrange(YearMonth)

cat("Monthly observations:", nrow(monthly_layoffs), "\n")
```

```
## Monthly observations: 66
```

```r
cat("Range:", as.character(min(monthly_layoffs$YearMonth)),
    "to", as.character(max(monthly_layoffs$YearMonth)), "\n")
```

```
## Range: 2020-03-01 to 2025-12-01
```

2.2 Convert to ts Object

```r
start_year  <- year(min(monthly_layoffs$YearMonth))
start_month <- month(min(monthly_layoffs$YearMonth))

layoffs_ts <- ts(monthly_layoffs$Total_Laid_Off,
                 start     = c(start_year, start_month),
```

```r
                 frequency = 12)

cat("ts object: start =", tsp(layoffs_ts)[1],
    "| end =", tsp(layoffs_ts)[2],
    "| length =", length(layoffs_ts), "\n")
```

```
## ts object: start = 2020.167 | end = 2025.583 | length = 66
```

2.3 Train / Test Split

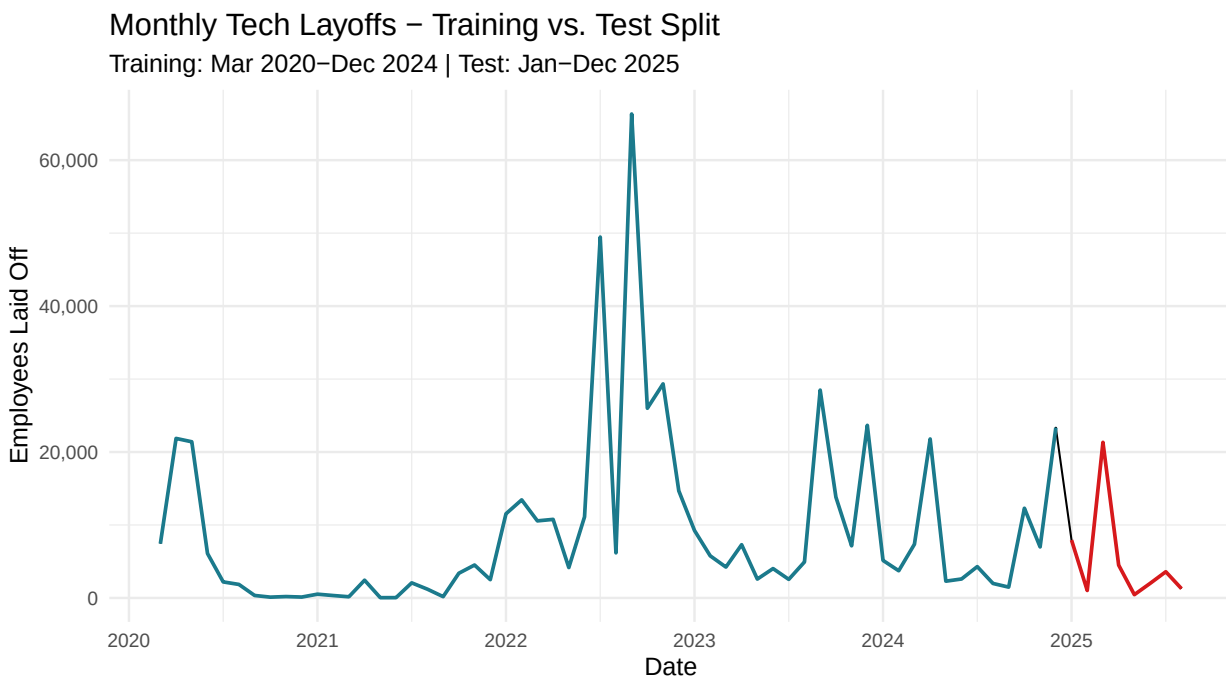
```r
train_ts <- window(layoffs_ts, end = c(2024, 12))

test_ts  <- window(layoffs_ts, start = c(2025,  1), end = c(2025, 12))

cat("Training observations:", length(train_ts), "\n")
```

```
## Training observations: 58
```

```r
cat("Test observations:    ", length(test_ts),  "\n")
```

```
## Test observations:     8
```

```r
autoplot(layoffs_ts) +
  autolayer(train_ts, series = "Training (2020-2024)", color = "#1B7A8C", linewidth = 0.9) +
  autolayer(test_ts,  series = "Test (2025)",          color = "#d7191c", linewidth = 0.9) +
  scale_y_continuous(labels = comma) +
  labs(
    title    = "Monthly Tech Layoffs - Training vs. Test Split",
    subtitle = "Training: Mar 2020-Dec 2024 | Test: Jan-Dec 2025",
    x = "Date", y = "Employees Laid Off", color = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```



---

3 Stationarity Confirmation

```
adf_train <- adf.test(train_ts, alternative = "stationary")
cat("ADF Test - Training Series (original)\n")
```

```
## ADF Test - Training Series (original)
```

```
cat("  Statistic:", round(adf_train$statistic, 3),
    " | p-value:", round(adf_train$p.value, 4), "\n")
```

```
##   Statistic: -2.889  | p-value: 0.2156
```

```
cat("  Conclusion:", ifelse(adf_train$p.value < 0.05, "Stationary", "Non-stationary"), "\n\n")
```

```
##   Conclusion: Non-stationary
```

```
train_diff <- diff(train_ts, differences = 1)
adf_diff   <- adf.test(train_diff, alternative = "stationary")
cat("ADF Test - Training Series (first-differenced)\n")
```

```
## ADF Test - Training Series (first-differenced)
```

```r
cat("  Statistic:", round(adf_diff$statistic, 3),
    " | p-value:", round(adf_diff$p.value, 4), "\n")
```

```
##   Statistic: -3.721  | p-value: 0.0309
```

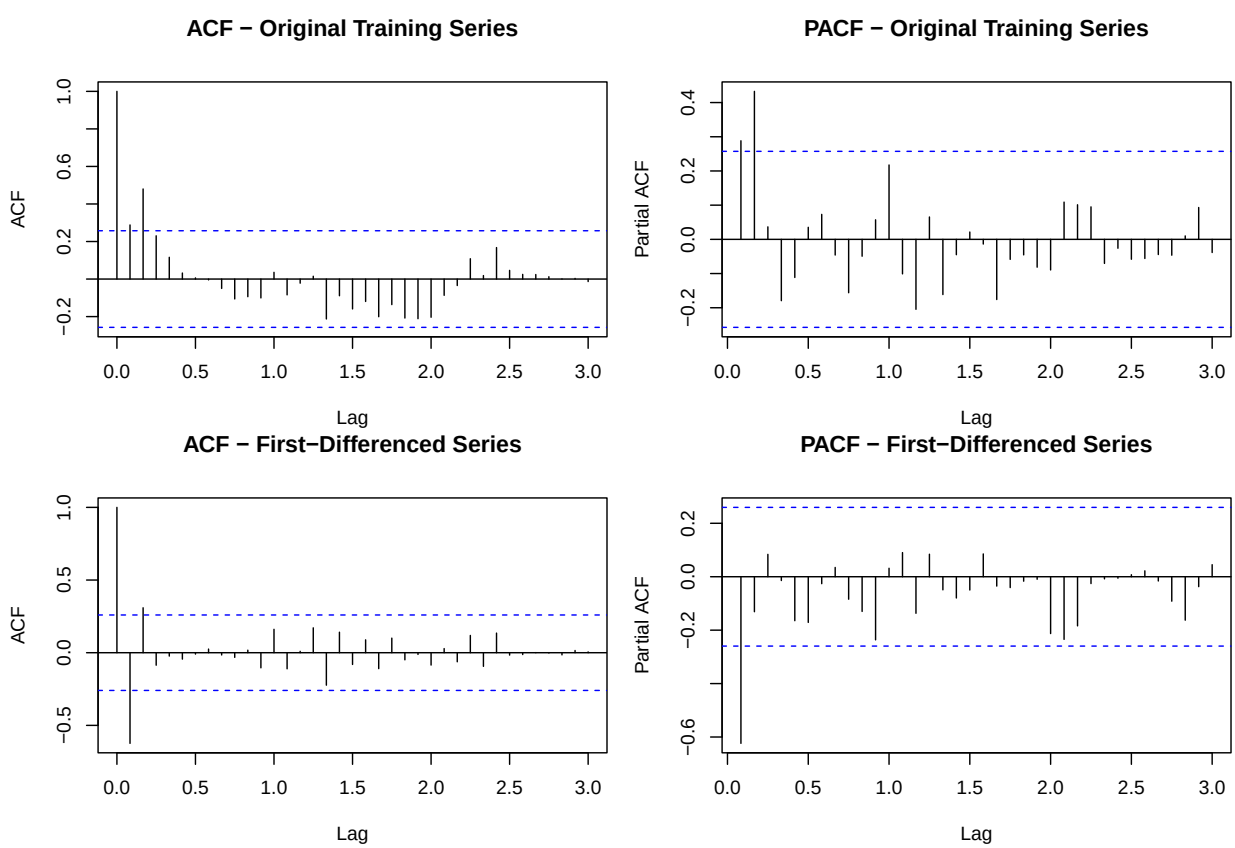
```r
cat("  Conclusion:", ifelse(adf_diff$p.value < 0.05, "Stationary", "Still non-stationary"), "\n")
```

```
##   Conclusion: Stationary
```

---

4 ACF and PACF Plots

```r
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
acf(train_ts,   lag.max = 36, main = "ACF - Original Training Series")
pacf(train_ts,  lag.max = 36, main = "PACF - Original Training Series")
acf(train_diff, lag.max = 36, main = "ACF - First-Differenced Series")
pacf(train_diff, lag.max = 36, main = "PACF - First-Differenced Series")
```



```r
par(mfrow = c(1, 1))
```

---

5 2023 Q1 Structural Break

```r
all_dates <- seq(as.Date("2020-03-01"), by = "month",
                 length.out = length(layoffs_ts))

xreg_full <- as.numeric(
  all_dates >= as.Date("2023-01-01") & all_dates <= as.Date("2023-03-01")
)

n_train <- length(train_ts)
xreg_train <- xreg_full[1:n_train]
xreg_test  <- xreg_full[(n_train + 1):(n_train + length(test_ts))]

cat("Break indicator in training (sum = months flagged):", sum(xreg_train), "\n")
```

```
## Break indicator in training (sum = months flagged): 3
```

```r
cat("Break indicator in test (should be 0):", sum(xreg_test), "\n")
```

```
## Break indicator in test (should be 0): 0
```

---

6 Model Fitting

6.1 Seasonal Naive Baseline

```r
h_test      <- length(test_ts)
snaive_fit  <- snaive(train_ts, h = h_test)
snaive_rmse <- sqrt(mean((snaive_fit$mean - test_ts)^2))
cat("Seasonal Naive RMSE:", round(snaive_rmse, 2), "\n")
```

```
## Seasonal Naive RMSE: 8013.9
```

6.2 auto.arima

```r
set.seed(2002)
model_auto <- auto.arima(
  train_ts,
  stepwise      = FALSE,
  approximation = FALSE,
  seasonal      = TRUE,
  ic            = "aicc",
  trace         = FALSE
)

summary(model_auto)
```

```
## Series: train_ts
## ARIMA(0,0,3) with non-zero mean
##
## Coefficients:
##          ma1     ma2     ma3      mean
##       0.0218  0.6732  0.4455  9580.208
## s.e.  0.1083  0.0868  0.1164  2592.223
##
## sigma^2 = 95332831:  log likelihood = -614.54
## AIC=1239.09   AICc=1240.24   BIC=1249.39
##
## Training set error measures:
##                     ME     RMSE      MAE       MPE     MAPE     MASE       ACF1
## Training set -40.18689 9421.155 6014.738 -882.5915 931.0008 0.533916 0.05969579
```

6.3 SARIMA(1,1,1)(1,1,0)[12]

```r
model_sarima <- Arima(
  train_ts,
  order          = c(1, 1, 1),
  seasonal       = list(order = c(1, 1, 0), period = 12)
)

summary(model_sarima)
```

```
## Series: train_ts
## ARIMA(1,1,1)(1,1,0)[12]
##
## Coefficients:
##           ar1      ma1     sar1
##       -0.5878  -0.0903  -0.2223
## s.e.   0.1798   0.2242   0.1414
##
## sigma^2 = 2.28e+08:  log likelihood = -495.88
## AIC=999.75   AICc=1000.75   BIC=1006.98
##
## Training set error measures:
##                   ME     RMSE      MAE      MPE     MAPE      MASE         ACF1
## Training set 305.836 12848.16 8217.541 51.46939 1146.969 0.7294543 -0.006737135
```

6.4 ARIMA(2,1,1)

```r
model_arima211 <- Arima(
  train_ts,
  order = c(2, 1, 1)
)

summary(model_arima211)
```

```
## Series: train_ts
## ARIMA(2,1,1)
##
## Coefficients:
##           ar1      ar2     ma1
##       -1.0333  -0.3477  0.2974
## s.e.   0.5051   0.3216  0.5223
##
## sigma^2 = 128585664:  log likelihood = -611.79
## AIC=1231.58   AICc=1232.35   BIC=1239.75
##
## Training set error measures:
##                    ME     RMSE      MAE      MPE     MAPE      MASE       ACF1
## Training set 202.5548 10941.56 6979.571 -194.545 239.8869 0.6195623 0.01259533
```

6.5 SARIMAX auto.arima

```r
set.seed(2002)
model_sarimax <- auto.arima(
  train_ts,
  xreg          = xreg_train,
  stepwise      = FALSE,
  approximation = FALSE,
  seasonal      = TRUE,
  ic            = "aicc",
  trace         = FALSE
)

summary(model_sarimax)
```

```
## Series: train_ts
## Regression with ARIMA(2,0,0) errors
##
## Coefficients:
##          ar1     ar2  intercept       xreg
##       0.1534  0.4499  10083.000  -6330.682
## s.e.  0.1162  0.1172   3310.206   6688.720
##
## sigma^2 = 115477837:  log likelihood = -618.87
## AIC=1247.73   AICc=1248.89   BIC=1258.03
##
```

```
## Training set error measures:
##                     ME     RMSE      MAE       MPE     MAPE     MASE
## Training set -76.18399 10368.89 6945.063 -802.8876 828.2886 0.616499
##                     ACF1
## Training set -0.01924922
```

---

7 AIC / BIC Comparison

```r
model_names <- c(
  "auto.arima (no xreg)",
  "SARIMA(1,1,1)(1,1,0)[12]",
  "ARIMA(2,1,1)",
  "SARIMAX (auto + break_2023)"
)

model_list <- list(model_auto, model_sarima, model_arima211, model_sarimax)

info_table <- tibble(
  Model   = model_names,
  AIC     = sapply(model_list, AIC),
  AICc    = sapply(model_list, function(m) m$aicc),
  BIC     = sapply(model_list, BIC),
  `Log-Lik` = sapply(model_list, logLik) %>% as.numeric()
) %>%
  arrange(AICc)

kable(info_table, digits = 2, caption = "Model Information Criteria Comparison") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE) %>%
  row_spec(1, bold = TRUE, background = "#d4edda")
```

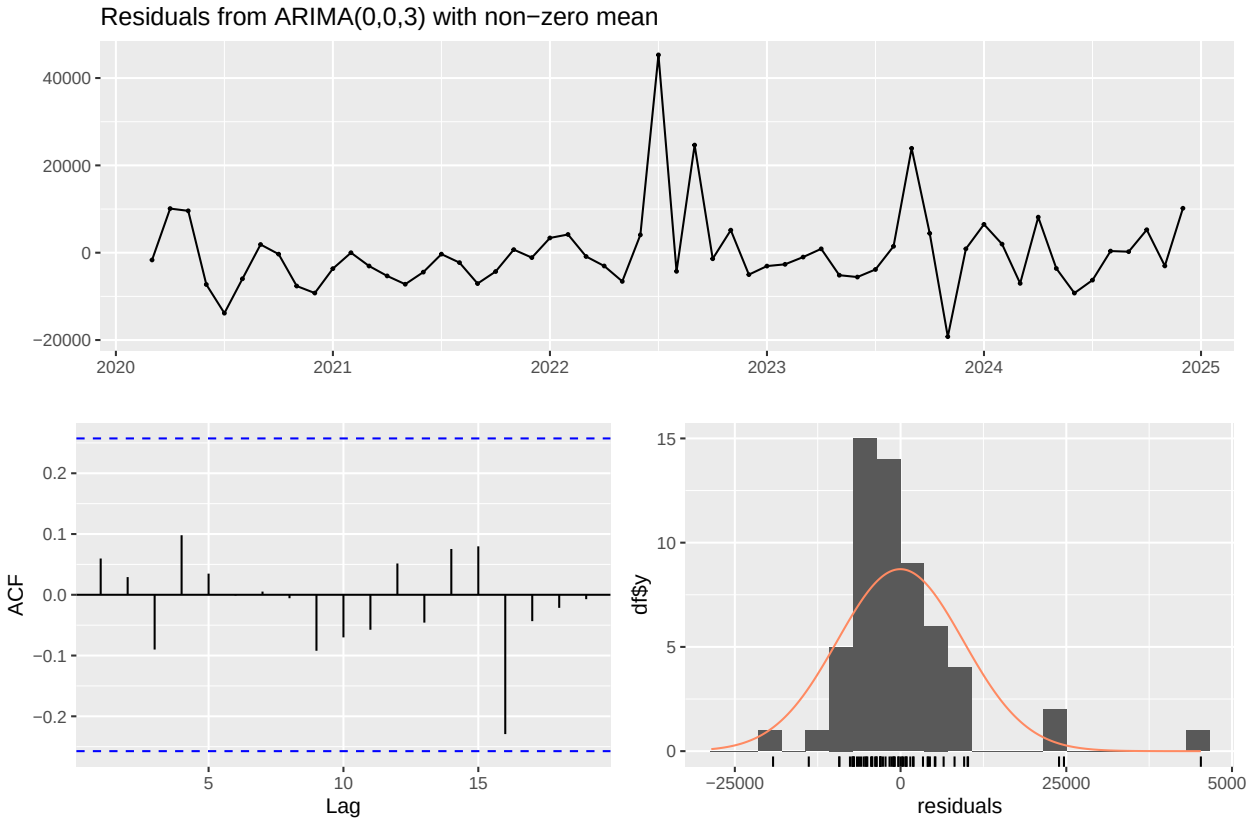
Table 1: Model Information Criteria Comparison

Model	AIC	AICc	BIC	Log-Lik
SARIMA(1,1,1)(1,1,0)[12]	**999.75**	**1000.75**	**1006.98**	**-495.88**
ARIMA(2,1,1)	1231.58	1232.35	1239.75	-611.79
auto.arima (no xreg)	1239.09	1240.24	1249.39	-614.54
SARIMAX (auto + break_2023)	1247.73	1248.89	1258.03	-618.87

---

8 Residual Diagnostics

```r
checkresiduals(model_auto)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,3) with non-zero mean
## Q* = 2.8924, df = 9, p-value = 0.9684
##
## Model df: 3.   Total lags used: 12
```

```
lb_auto <- Box.test(residuals(model_auto), lag = 12, type = "Ljung-Box", fitdf = length(coef(model_auto
cat("Ljung-Box Test (auto.arima)\n")
```

```
## Ljung-Box Test (auto.arima)
```

```
cat("  Chi-squared:", round(lb_auto$statistic, 3), " | df:", lb_auto$parameter,
    " | p-value:", round(lb_auto$p.value, 4), "\n")
```

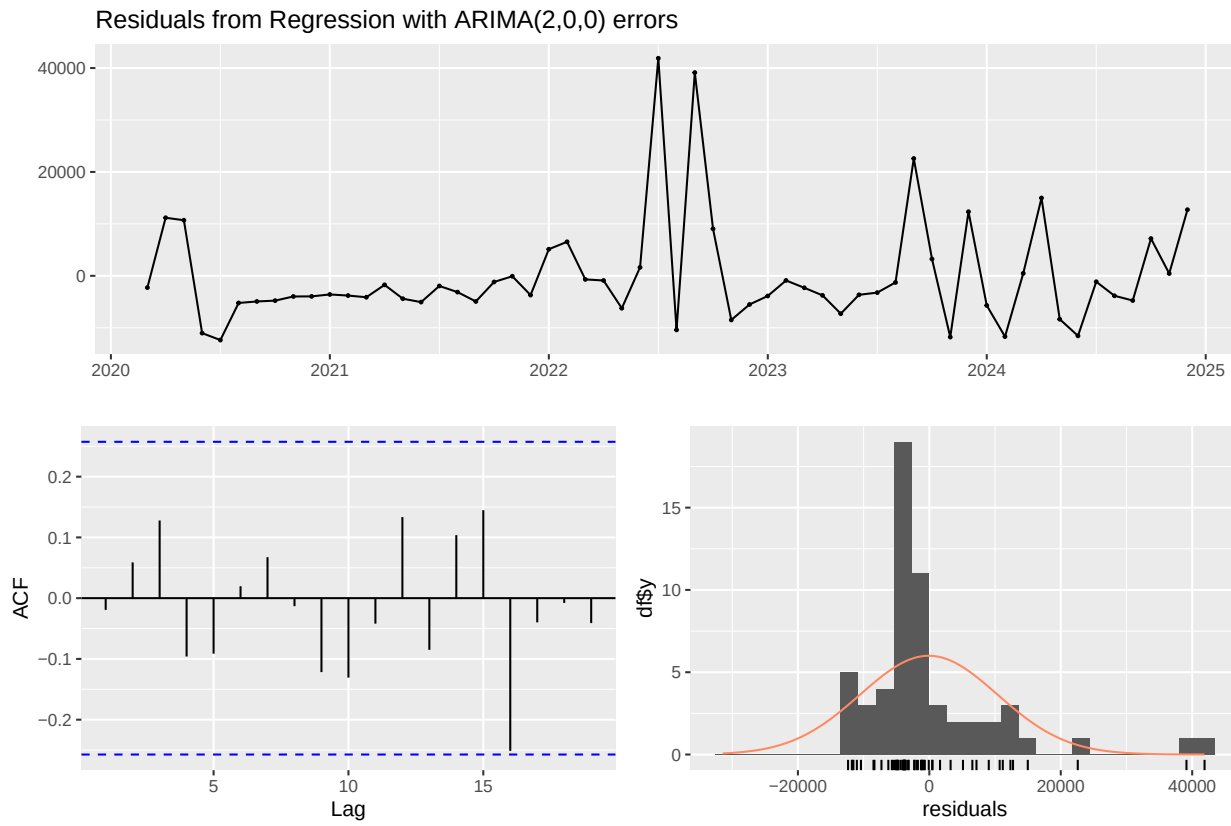
```
##   Chi-squared: 2.892  | df: 8  | p-value: 0.9409
```

```
cat("  Conclusion:", ifelse(lb_auto$p.value > 0.05,
                            "Residuals are white noise (good)",
                            "Significant autocorrelation remains in residuals"), "\n")
```

```
##   Conclusion: Residuals are white noise (good)
```

8.1 SARIMAX Model

```r
checkresiduals(model_sarimax)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from Regression with ARIMA(2,0,0) errors
## Q* = 6.5309, df = 10, p-value = 0.7689
##
## Model df: 2.   Total lags used: 12
```

```r
lb_sarimax <- Box.test(residuals(model_sarimax), lag = 12, type = "Ljung-Box",
                       fitdf = length(coef(model_sarimax)))
cat("Ljung-Box Test (SARIMAX)\n")
```

```
## Ljung-Box Test (SARIMAX)
```

```r
cat("  Chi-squared:", round(lb_sarimax$statistic, 3), " | df:", lb_sarimax$parameter,
    " | p-value:", round(lb_sarimax$p.value, 4), "\n")
```

```
##   Chi-squared: 6.531  | df: 8  | p-value: 0.588
```

```r
cat("  Conclusion:", ifelse(lb_sarimax$p.value > 0.05,
                            "Residuals are white noise (good)",
                            "Significant autocorrelation remains in residuals"), "\n")
```

```
##   Conclusion: Residuals are white noise (good)
```

---

9 Evaluation on Test Set

```r
fc_snaive   <- forecast(snaive_fit,    h = h_test)
fc_auto     <- forecast(model_auto,    h = h_test)
fc_sarima   <- forecast(model_sarima,  h = h_test)
fc_arima211 <- forecast(model_arima211, h = h_test)
fc_sarimax  <- forecast(model_sarimax, h = h_test, xreg = xreg_test)
```

```r
rmse <- function(actual, predicted) sqrt(mean((actual - predicted)^2))
mae  <- function(actual, predicted) mean(abs(actual - predicted))
mape <- function(actual, predicted) mean(abs((actual - predicted) / actual)) * 100

eval_table <- tibble(
  Model = c(
    "Seasonal Naive (Baseline)",
    "auto.arima (no xreg)",
    "SARIMA(1,1,1)(1,1,0)[12]",
    "ARIMA(2,1,1)",
    "SARIMAX (auto + break_2023)"
  ),
  RMSE = c(
    rmse(test_ts, fc_snaive$mean),
    rmse(test_ts, fc_auto$mean),
    rmse(test_ts, fc_sarima$mean),
    rmse(test_ts, fc_arima211$mean),
    rmse(test_ts, fc_sarimax$mean)
  ),
  MAE = c(
    mae(test_ts, fc_snaive$mean),
    mae(test_ts, fc_auto$mean),
    mae(test_ts, fc_sarima$mean),
    mae(test_ts, fc_arima211$mean),
    mae(test_ts, fc_sarimax$mean)
  ),
  MAPE = c(
    mape(test_ts, fc_snaive$mean),
    mape(test_ts, fc_auto$mean),
    mape(test_ts, fc_sarima$mean),
    mape(test_ts, fc_arima211$mean),
    mape(test_ts, fc_sarimax$mean)
  )
) %>%
```

```r
  arrange(RMSE)

kable(eval_table, digits = 2,
      caption = "Out-of-Sample Forecast Accuracy on Test Set (Jan-Dec 2025)") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE) %>%
  row_spec(1, bold = TRUE, background = "#d4edda") %>%
  row_spec(which(eval_table$Model == "Seasonal Naive (Baseline)"),
           bold = FALSE, background = "#fff3cd")
```

Table 2: Out-of-Sample Forecast Accuracy on Test Set (Jan–Dec 2025)

Model	RMSE	MAE	MAPE
SARIMA(1,1,1)(1,1,0)[12]	**7745.41**	**4912.28**	**107.44**
Seasonal Naive (Baseline)	8013.90	5076.88	158.03
auto.arima (no xreg)	8108.46	7442.32	596.11
SARIMAX (auto + break_2023)	9723.23	9169.66	691.55
ARIMA(2,1,1)	12883.02	12087.93	980.92

```r
best_rmse    <- min(eval_table$RMSE[eval_table$Model != "Seasonal Naive (Baseline)"])
baseline_rmse <- eval_table$RMSE[eval_table$Model == "Seasonal Naive (Baseline)"]
best_model   <- eval_table$Model[eval_table$RMSE == best_rmse]

cat("=== QUANTIFIABLE GOAL CHECK ===\n")
```

```
## === QUANTIFIABLE GOAL CHECK ===
```

```r
cat("Seasonal Naive RMSE:", round(baseline_rmse, 2), "\n")
```

```
## Seasonal Naive RMSE: 8013.9
```

```r
cat("Best model RMSE:    ", round(best_rmse,    2), "(", best_model, ")\n")
```

```
## Best model RMSE:     7745.41 ( SARIMA(1,1,1)(1,1,0)[12] )
```

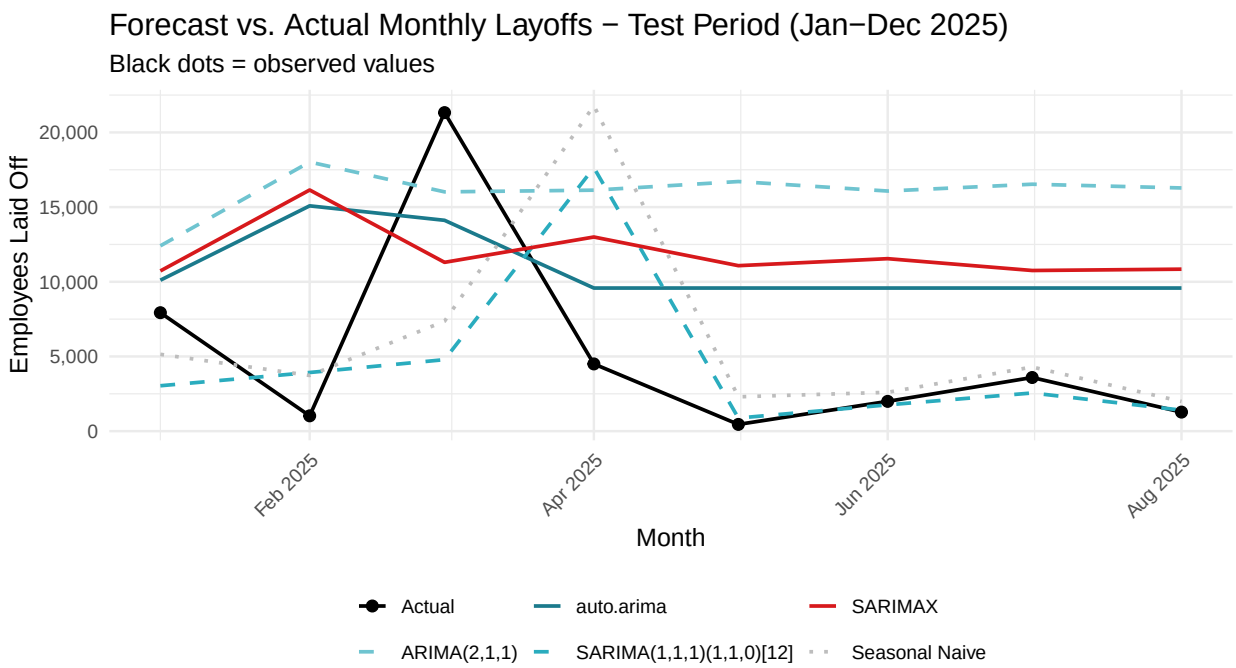
```r
cat("Goal achieved:      ",
    ifelse(best_rmse < baseline_rmse, "YES - Beat the baseline!", "NO - Did not beat baseline."), "\n")
```

```
## Goal achieved:       YES - Beat the baseline!
```

9.1 Forecast vs. Actuals Plot

```r
test_dates <- seq(as.Date("2025-01-01"), by = "month", length.out = h_test)

fc_df <- tibble(
  Date    = test_dates,
```

```r
  Actual  = as.numeric(test_ts),
  `Seasonal Naive`            = as.numeric(fc_snaive$mean)[1:h_test],
  `auto.arima`                = as.numeric(fc_auto$mean)[1:h_test],
  `SARIMA(1,1,1)(1,1,0)[12]` = as.numeric(fc_sarima$mean)[1:h_test],
  `ARIMA(2,1,1)`              = as.numeric(fc_arima211$mean)[1:h_test],
  `SARIMAX`                   = as.numeric(fc_sarimax$mean)[1:h_test]
) %>%
  pivot_longer(-Date, names_to = "Series", values_to = "Layoffs")

ggplot(fc_df, aes(x = Date, y = Layoffs, color = Series, linetype = Series)) +
  geom_line(linewidth = 0.9) +
  geom_point(data = filter(fc_df, Series == "Actual"), size = 2.5) +
  scale_y_continuous(labels = comma) +
  scale_x_date(date_breaks = "2 months", date_labels = "%b %Y") +
  scale_color_manual(values = c(
    "Actual"                   = "#000000",
    "Seasonal Naive"           = "#bdbdbd",
    "auto.arima"               = "#1B7A8C",
    "SARIMA(1,1,1)(1,1,0)[12]"= "#2AACBE",
    "ARIMA(2,1,1)"             = "#6FC4D0",
    "SARIMAX"                  = "#d7191c"
  )) +
  scale_linetype_manual(values = c(
    "Actual"                   = "solid",
    "Seasonal Naive"           = "dotted",
    "auto.arima"               = "solid",
    "SARIMA(1,1,1)(1,1,0)[12]"= "dashed",
    "ARIMA(2,1,1)"             = "dashed",
    "SARIMAX"                  = "solid"
  )) +
  labs(
    title    = "Forecast vs. Actual Monthly Layoffs - Test Period (Jan-Dec 2025)",
    subtitle = "Black dots = observed values",
    x = "Month", y = "Employees Laid Off", color = NULL, linetype = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 45, hjust = 1))
```

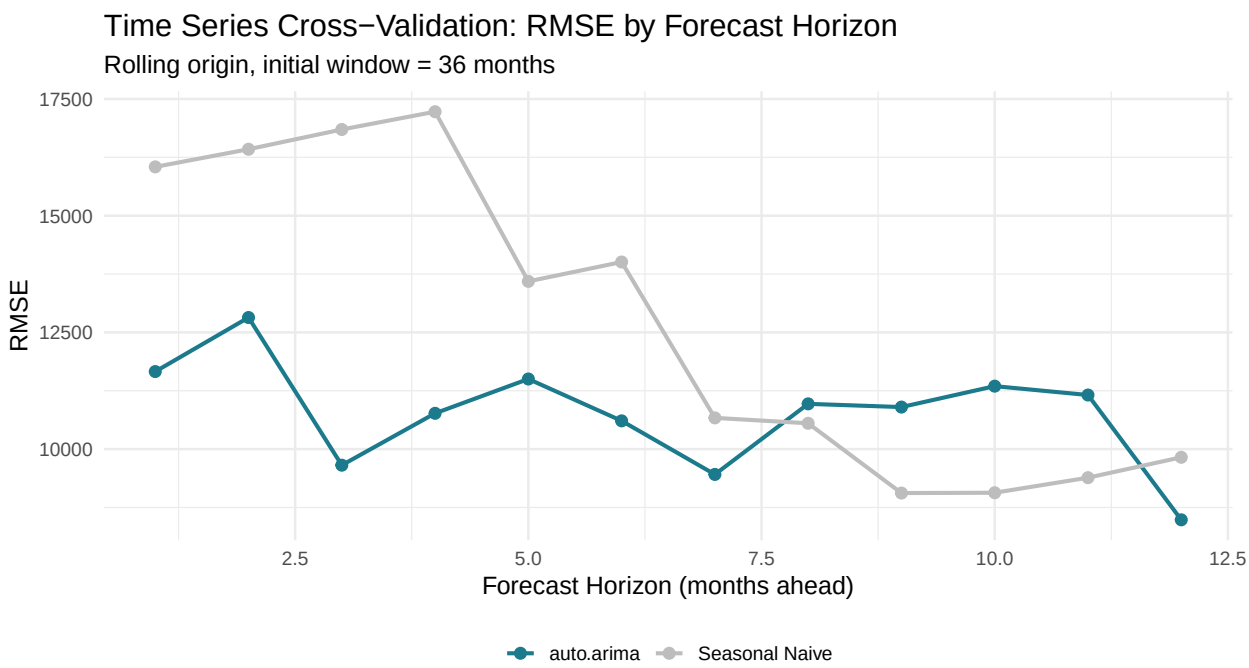


---

10 CrossnValidation

```
fc_auto_fn <- function(y, h) {
  forecast(auto.arima(y, stepwise = TRUE, approximation = TRUE, seasonal = TRUE), h = h)
}

fc_snaive_fn <- function(y, h) snaive(y, h = h)

cv_auto   <- tsCV(train_ts, fc_auto_fn,   h = 12, initial = 36)
cv_snaive <- tsCV(train_ts, fc_snaive_fn, h = 12, initial = 36)

cv_rmse_auto   <- sqrt(colMeans(cv_auto^2,   na.rm = TRUE))
cv_rmse_snaive <- sqrt(colMeans(cv_snaive^2, na.rm = TRUE))

cv_plot_df <- tibble(
  Horizon      = 1:12,
  `auto.arima` = cv_rmse_auto,
  `Seasonal Naive` = cv_rmse_snaive
) %>%
  pivot_longer(-Horizon, names_to = "Model", values_to = "RMSE")

ggplot(cv_plot_df, aes(x = Horizon, y = RMSE, color = Model)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2.5) +
  scale_color_manual(values = c("auto.arima" = "#1B7A8C", "Seasonal Naive" = "#bdbdbd")) +
  labs(
```

```r
    title    = "Time Series Cross-Validation: RMSE by Forecast Horizon",
    subtitle = "Rolling origin, initial window = 36 months",
    x = "Forecast Horizon (months ahead)", y = "RMSE", color = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom")
```




```r
cat("Mean CV RMSE - auto.arima:    ", round(mean(cv_rmse_auto,   na.rm = TRUE), 2), "\n")
```

```
## Mean CV RMSE - auto.arima:     10777.47
```

```r
cat("Mean CV RMSE - Seasonal Naive:", round(mean(cv_rmse_snaive, na.rm = TRUE), 2), "\n")
```

```
## Mean CV RMSE - Seasonal Naive: 12724.98
```

```r
best_model_obj <- switch(
  best_model,
  "auto.arima (no xreg)"        = model_auto,
  "SARIMA(1,1,1)(1,1,0)[12]"    = model_sarima,
  "ARIMA(2,1,1)"                = model_arima211,
  "SARIMAX (auto + break_2023)" = model_sarimax,
  model_auto  # fallback
)

cat("=== BEST MODEL ===\n")
```

```
## === BEST MODEL ===
```

```r
print(best_model_obj)
```

```
## Series: train_ts
## ARIMA(1,1,1)(1,1,0)[12]
##
## Coefficients:
##           ar1      ma1     sar1
##       -0.5878  -0.0903  -0.2223
## s.e.   0.1798   0.2242   0.1414
##
## sigma^2 = 2.28e+08:  log likelihood = -495.88
## AIC=999.75   AICc=1000.75   BIC=1006.98
```

---

11 Forecast: 2026–2027

```r
full_xreg <- rep(0, length(layoffs_ts))

model_final <- auto.arima(
  layoffs_ts,
  stepwise      = FALSE,
  approximation = FALSE,
  seasonal      = TRUE,
  ic            = "aicc",
  trace         = FALSE
)

cat("Final model (fit on full data):\n")
```

```
## Final model (fit on full data):
```

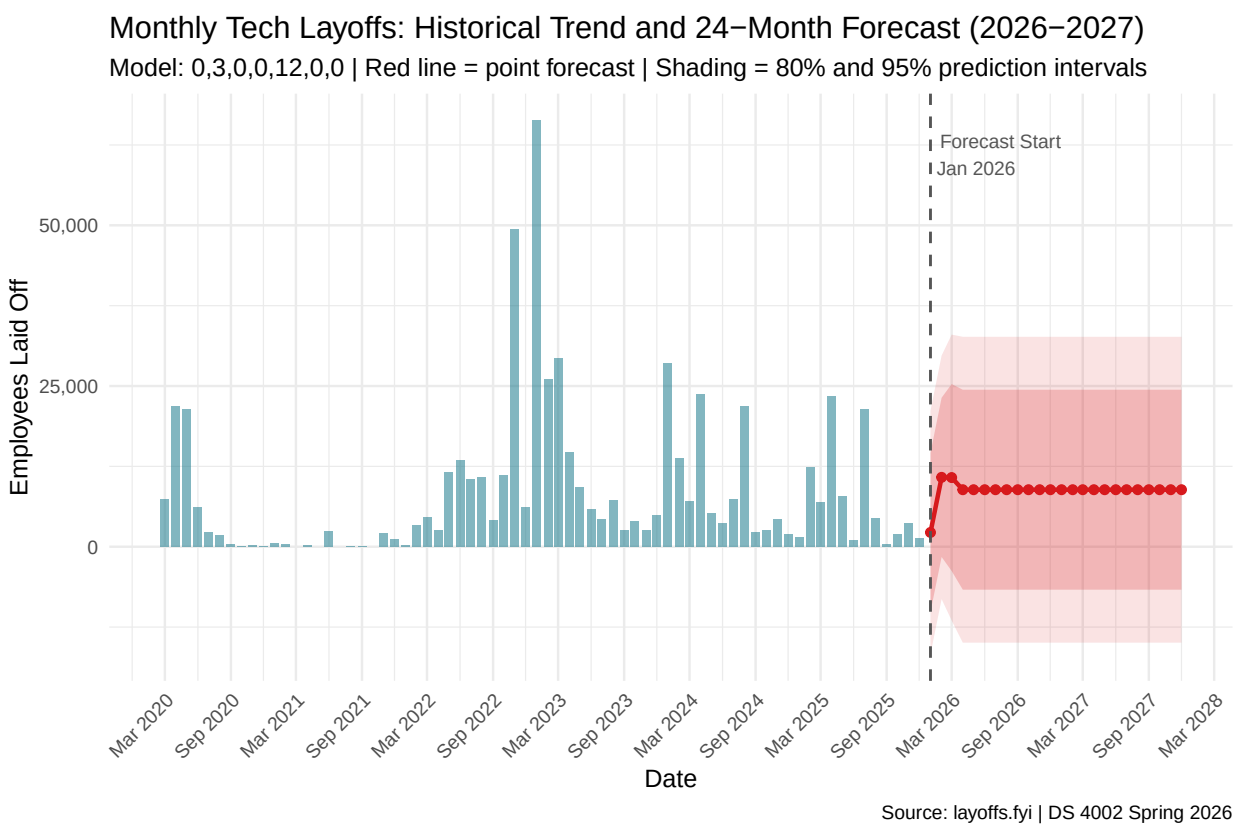
```r
print(model_final)
```

```
## Series: layoffs_ts
## ARIMA(0,0,3) with non-zero mean
##
## Coefficients:
##          ma1     ma2     ma3      mean
##       0.0356  0.6153  0.4464  8867.620
## s.e.  0.1064  0.0801  0.1084  2373.957
##
## sigma^2 = 93308044:  log likelihood = -698.35
## AIC=1406.71   AICc=1407.71   BIC=1417.66
```

```r
h_future <- 24
fc_final  <- forecast(model_final, h = h_future, level = c(80, 95))

forecast_dates <- seq(as.Date("2026-01-01"), by = "month", length.out = h_future)

fc_future_df <- tibble(
  Date     = forecast_dates,
  Forecast = as.numeric(fc_final$mean),
  Lo80     = as.numeric(fc_final$lower[, 1]),
  Hi80     = as.numeric(fc_final$upper[, 1]),
  Lo95     = as.numeric(fc_final$lower[, 2]),
  Hi95     = as.numeric(fc_final$upper[, 2])
)

hist_df <- tibble(
  Date    = monthly_layoffs$YearMonth,
  Layoffs = monthly_layoffs$Total_Laid_Off
)
```

```r
ggplot() +

  geom_col(data = hist_df, aes(x = Date, y = Layoffs),
           fill = "#1B7A8C", alpha = 0.55, width = 25) +

  geom_ribbon(data = fc_future_df,
              aes(x = Date, ymin = Lo95, ymax = Hi95),
              fill = "#d7191c", alpha = 0.12) +

  geom_ribbon(data = fc_future_df,
              aes(x = Date, ymin = Lo80, ymax = Hi80),
              fill = "#d7191c", alpha = 0.22) +

  geom_line(data = fc_future_df,
            aes(x = Date, y = Forecast),
            color = "#d7191c", linewidth = 1.1) +
  geom_point(data = fc_future_df,
             aes(x = Date, y = Forecast),
             color = "#d7191c", size = 2) +

  geom_vline(xintercept = as.Date("2026-01-01"),
             linetype = "dashed", color = "#555555", linewidth = 0.7) +
  annotate("text", x = as.Date("2026-01-01"), y = max(hist_df$Layoffs) * 0.92,
           label = "Forecast Start\nJan 2026", hjust = -0.08,
           size = 3.5, color = "#555555") +
  scale_y_continuous(labels = comma) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  labs(
    title    = "Monthly Tech Layoffs: Historical Trend and 24-Month Forecast (2026-2027)",
    subtitle = paste0("Model: ", model_final$arma |> paste(collapse=","),
                      " | Red line = point forecast | Shading = 80% and 95% prediction intervals"),
    x = "Date", y = "Employees Laid Off",
    caption = "Source: layoffs.fyi | DS 4002 Spring 2026"
  ) +
```

```
  theme_minimal(base_size = 13) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



11.1 Forecast Table

```
fc_future_df %>%
  mutate(
    Year  = year(Date),
    Month = month(Date, label = TRUE)
  ) %>%
  select(Year, Month, Forecast, Lo80, Hi80, Lo95, Hi95) %>%
  mutate(across(Forecast:Hi95, ~ comma(round(.x)))) %>%
  kable(caption = "24-Month Forecast: Monthly Tech Layoffs (Jan 2026 - Dec 2027)") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 3: 24-Month Forecast: Monthly Tech Layoffs (Jan 2026 – Dec 2027)

Year	Month	Forecast	Lo80	Hi80	Lo95	Hi95
2026	Jan	2,221	-10,158	14,601	-16,711	21,154
2026	Feb	10,794	-1,594	23,181	-8,151	29,738

2026	Mar	10,759	-3,782	25,301	-11,480	32,999
2026	Apr	8,868	-6,689	24,424	-14,924	32,659
2026	May	8,868	-6,689	24,424	-14,924	32,659
2026	Jun	8,868	-6,689	24,424	-14,924	32,659
2026	Jul	8,868	-6,689	24,424	-14,924	32,659
2026	Aug	8,868	-6,689	24,424	-14,924	32,659
2026	Sep	8,868	-6,689	24,424	-14,924	32,659
2026	Oct	8,868	-6,689	24,424	-14,924	32,659
2026	Nov	8,868	-6,689	24,424	-14,924	32,659
2026	Dec	8,868	-6,689	24,424	-14,924	32,659
2027	Jan	8,868	-6,689	24,424	-14,924	32,659
2027	Feb	8,868	-6,689	24,424	-14,924	32,659
2027	Mar	8,868	-6,689	24,424	-14,924	32,659
2027	Apr	8,868	-6,689	24,424	-14,924	32,659
2027	May	8,868	-6,689	24,424	-14,924	32,659
2027	Jun	8,868	-6,689	24,424	-14,924	32,659
2027	Jul	8,868	-6,689	24,424	-14,924	32,659
2027	Aug	8,868	-6,689	24,424	-14,924	32,659
2027	Sep	8,868	-6,689	24,424	-14,924	32,659
2027	Oct	8,868	-6,689	24,424	-14,924	32,659
2027	Nov	8,868	-6,689	24,424	-14,924	32,659
2027	Dec	8,868	-6,689	24,424	-14,924	32,659

Table 4: Summary of Modeling Results

Item	Result
Training period	Mar 2020 – Dec 2024
Test period	Jan 2025 – Aug 2025
Forecast horizon	Jan 2026 – Dec 2027 (24 months)
Seasonal naive RMSE (baseline)	8,014
Best model RMSE (SARIMA(1,1,1)(1,1,0)[12])	7,745
Goal achieved (beat baseline)?	YES
ADF p-value (original series)	0.2156
ADF p-value (differenced series)	0.0309
Ljung-Box p-value (best model)	0.9409